SAHIL CHHABRA

B.Tech Electronics Engineering – VLSI Design & Technology | Vivekananda Institute of Professional Studies – TC | 2024–2028

Delhi, India • +91 93557 98225 • • Portfolio • Open to Internships

PROFESSIONAL SUMMARY

Electronics Engineering undergraduate specializing in analog and mixed-signal IC design, with hands-on full-custom layout, schematic, and SPICE-level verification experience in Cadence Virtuoso. Built and DRC/LVS-verified layouts for a buck converter, transconductance amplifier, and 6T SRAM cell, and designed low-power adiabatic-logic ALUs verified through PVT analysis across voltage variation, a wide temperature range, and all major process corners. Authored a published, IEEE-presented paper on a temperature-robust ALU for Quantum inspired computation architectures and a submitted journal paper on a PVT-insensitive energy-recovering ALU for different Microcontroller and Microprocessor ; currently working on PLL design. Complements analog design skills with a full RTL-to-GDSII digital flow background.

TECHNICAL SKILLS

VLSI / IC Design	Analog & Mixed-Signal IC Design • Full-Custom Layout Design • Low-Power / Adiabatic Logic • Memory (SRAM) Architecture Design • PLL • Digital Design & Verification
EDA Tools	Cadence (Virtuoso, Genus, Innovus, Xcelium) • AMD Xilinx Vivado
Verification	DRC • LVS • Noise Margin / SNM (Butterfly-Curve) Analysis • PVT Analysis • Process-Corner Analysis (SS, FS, SF, TT, FF) • Monte Carlo Analysis • PSD Analysis • Timing Analysis
HDL / Programming	Verilog • C • Python • SQL • EMU8086 • Keil • MATLAB
Platforms	Windows 11 • Ubuntu • Red Hat Linux

RESEARCH INTERNSHIP

Research Intern, Dept. of Electronics & Communication Engineering – IIT Roorkee *Jun 2026 – Jul 2026*

Under the supervision of Prof. Sudeb Dasgupta

- Simulated and characterized an unsigned 8-bit Multiply-Accumulate (MAC) operation built on a 6T SRAM-based architecture in the digital domain, evaluating computational accuracy and hardware performance across varying operating conditions and benchmarking efficiency against conventional digital MAC implementations.

KEY PROJECTS — CADENCE VIRTUOSO

ThermaStable: Temperature-Resilient OTA for EV Battery Management Systems – Chip to Future (C2F) Hackathon – SoC Track *Hackathon*

- Co-designed (2-member team) a temperature- and voltage-insensitive Operational Transconductance Amplifier (OTA) in 180nm CMOS at 1.8V supply for BMS analog front-ends, built from a bandgap-based bias generation stage co-optimized with the amplification stage to remove gain drift at the circuit level.
- Achieved a gain error of just 1.18% (+25°C to +50°C) and 2.23% (+25°C to −40°C) with all transistors held in saturation; simulated in Cadence ADE L/ADE XL and implemented layout with Layout XL, including parasitic extraction.

PFD for High-Speed PLL Using Resettable D Flip-Flop & Charge Pump *5th Sem*

- Designed and verified a Phase-Frequency Detector built from a resettable D flip-flop and charge pump for high-speed PLL applications, verifying phase/frequency-error detection and lock behavior.

7-Stage Ring Oscillator Using Inverters *5th Sem*

- Designed and verified a 7-stage inverter-based ring oscillator, analyzing oscillation frequency and output signal stability.

Unsigned 8-Bit MAC Operation Using 6T SRAM in Digital Domain – IIT Roorkee Internship *Internship*

- Designed and verified an unsigned 8-bit MAC architecture using a 6T SRAM-based digital-domain implementation, validating functional correctness across multiple operating conditions.

Low-Power ALU Using Adiabatic Logic (Transmission Gates) *4th Sem*

- Designed a low-power ALU implementing AND/NAND gates with adiabatic transmission-gate logic, extended to a full 8-bit adiabatic ALU; verified via PVT analysis across ±5% voltage variation, −40°C–100°C, and SS/FS/SF/TT/FF corners.

Layout Design of Buck Converter & Transconductance Amplifier *4th Sem*

- Designed full-custom layouts in Cadence Virtuoso for a buck converter and a transconductance amplifier; verified manufacturability with DRC and LVS.

Adiabatic Logic Implementation – Logic Gates & Half Adder *4th Sem*

- Implemented adiabatic logic across fundamental gates and a half adder using transmission-gate topology, reducing dynamic power vs. conventional CMOS. [METRIC NEEDED – % power reduction]

RTL-to-GDS of a Half Adder *4th Sem*

- Executed the complete RTL-to-GDS flow (synthesis, placement, routing) for a half adder in Cadence, verified with DRC and LVS.

6T SRAM Cell with Static Noise Margin (SNM) Analysis *2nd Sem*

- Designed a 6T SRAM bitcell and characterized stability via butterfly-curve SNM analysis across hold/read/write modes under process and voltage variation. [METRIC NEEDED – SNM values in mV]

Schmitt Trigger Design & Analysis *2nd Sem*

- Designed and simulated a Schmitt trigger, analyzing hysteresis width, switching thresholds, and noise immunity.

RESEARCH PUBLICATIONS

Published (2026) — "Low-Temperature Power-Efficient Design of an 8-Bit ALU for Quantum-Inspired Computational Architectures"
Sahil Chhabra, Abhinav Sharma, Nishtha Kansal — 1st International Conference on VLSI Circuits and Systems (ICVCS 2026), IEEE CAS Kolkata Section, Jul 2026. Presented & published on IEEE Xplore.

- Designed a transistor-level 8-bit CMOS ALU integrating adiabatic logic; characterized performance across −40°C to 100°C (<1% variation) and validated robustness across all process corners and voltage fluctuations (<4% error margin).

Submitted — "PVT Insensitive Energy-Recovering ALU Architecture for Low-Power Microprocessor and Microcontroller Applications"
Sahil Chhabra, Abhinav Sharma, Nishtha Kansal — Manuscript submitted to a peer-reviewed journal.

- Developed an energy-recovering ALU using adiabatic logic engineered for PVT-insensitive operation, validated across ±5% voltage variation, −40°C–100°C, and SS/FS/SF/TT/FF process corners.

CERTIFICATIONS

- Certificate of Presentation — ICVCS 2026, IEEE CAS Kolkata Section, Jul 2026
- Author Certificate — ICVCS 2026, IEEE CAS Kolkata Section, Jul 2026
- Winter School of Electronics — VIPS-TC, 2025
- Value Added Course (VAC): Cadence-Powered Circuit Design — VIPS-TC, 2025
- Two-Day Workshop on VLSI Design and Semiconductor Industry — ChipXpert Technologies Pvt. Ltd., Apr 2026
- VLSI SoC Design using Verilog HDL (Seminar) — VLSI For All, with VIPS-TC, Apr 2026
- Analog IC Design Implementation and Verification v1.0 (Online) — Cadence Training Services, Jun 2026
- Digital Design and Verification v1.0 (Online) — Cadence Training Services, Jun 2026
- Cadence RTL-to-GDSII Flow v7.0 (Online) — Cadence Training Services, Jun 2026
- Virtuoso Schematic Editor S1: Creating Design Schematics vIC25.1 (Online) — Cadence Training Services, Jul 2026

HACKATHON PARTICIPATION

Chip to Future (C2F) National Hackathon – SoC Track – VIPS-TC · Team of 2 *2026*

- Competed as a 2-member team (with Abhinav Sharma) designing ThermaStable, a temperature- and voltage-stable OTA for EV battery-management system.

EDUCATION

B.Tech, Electronics Engineering – VLSI Design & Technology – VIPS-TC *2024 – 2028*

- GPA: 8.5 / 10 (till 4th Semester)

Class XII (Science Stream) – Vishal Bharti Sr. Sec. School *2012 – 2024*

- Scored 80%

LEADERSHIP & ACTIVITIES

Technical Head – Cadence & Public Relations (PR) – Chip Crafters, Official Society of VSET *Aug 2025 – Sep 2026*

- Led Cadence Virtuoso workshops and hands-on IC design sessions; organized VLSI seminars and design challenges.
- Handled society outreach and communication, coordinating promotion and participation for events.

LANGUAGES

English – Professional Proficiency • Hindi – Native